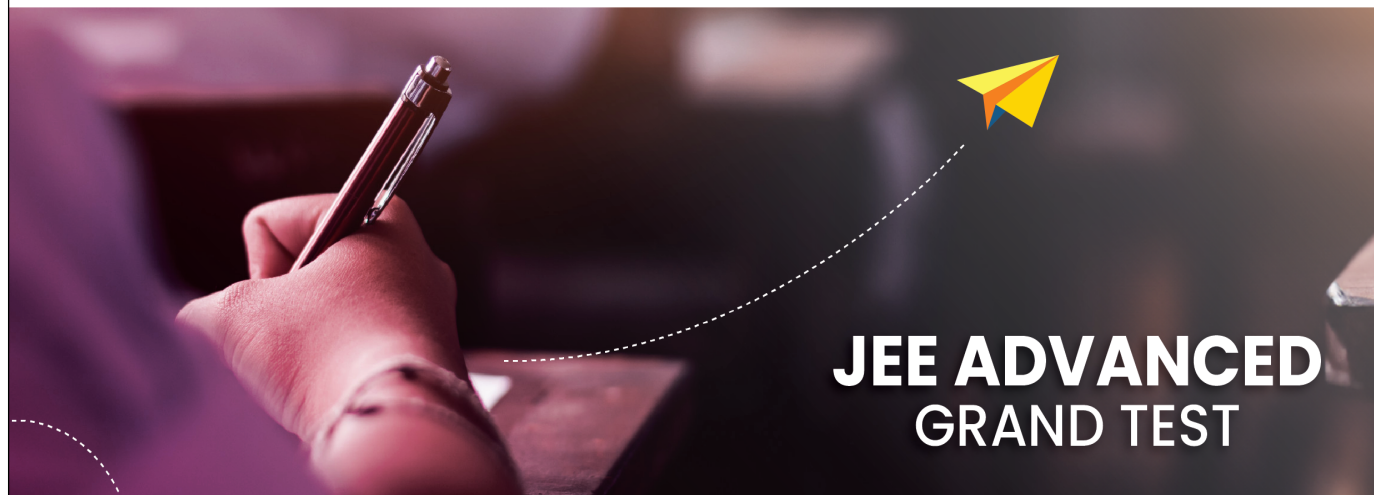




Sri Chaitanya
Educational Institutions



JEE ADVANCED GRAND TEST



Sri Chaitanya IIT Academy.,India.

✦ A.P ✦ T.S ✦ KARNATAKA ✦ TAMILNADU ✦ MAHARASTRA ✦ DELHI ✦ RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: **Sr.Super60_Elite, Target & LIIT-BTs**

Paper -1(Adv-2022-P1-Model)

Date: 12-05-2024

Time: 09.00Am to 12.00Pm

GTA-32

Max. Marks: 180

12-05-2024_**Sr.Super60_Elite, Target & LIIT-BTs**_Jee-Adv(2022-P1)_GTA-32_Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2022-P1-Model**

Time:3Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Matching Type	+3	-1	4	12
Total				18	60

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Matching Type	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Numerical Value Answer Type	+3	0	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Matching Type	+3	-1	4	12
Total				18	60

Sec: Sr.Super60_Elite, Target & LIIT-BTs

Space for rough work

Page 2

**Sri Chaitanya**
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**
RANK
1
SINGARAJU
VENKAT KOUNDINNYA
Sri Chaitanya
JEE 2nd Class
300
300



**JEE Advanced
2023**
RANK
1
VARIVILAS
CHIDOVILAS REDDY
Sri Chaitanya
JEE 2nd Class
341
360



**NEET
2023**
RANK
1
BORA VARUN
CHAKRAVARTHI
Sri Chaitanya
JEE 2nd Class
720
720



MATHEMATICS

Max Marks: 60

SECTION – I
(NUMERICAL VALUE TYPE)

This section contains EIGHT (08) questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;

Partial Mark: 0 In all other cases.

1. The number of solution of the equation $(\sin 3x - 1)(|\sqrt{3}\tan x + 1| + |2\cos 2x - 1|) = 0$ in the interval $[0, 100\pi]$ is
2. For positive integer m and n such that $m + 10 < n + 1$, both the mean and the median of the set $\{m, m + 4, m + 10, n + 1, n + 2, 2n\}$ are equal to n . What is $m + n$?
3. If the area of a square circumscribing an ellipse is A square units. The maximum distance of a normal from the centre of ellipse is r , then the value of $\left\lceil \frac{A}{14 \cdot r^2} \right\rceil$ is (Where $[\cdot]$ denotes greatest integer function) (Given eccentricity of ellipse is 0.5)
4. If $a, b, c \in N$, the probability that $a^2 + b^2 + c^2 = 13k, k \in N$ is $\frac{m}{n}$, where m, n are relatively prime numbers, then $(n - m)$ is
5. Let $S_1 = 0$ and $S_2 = 0$ be two circles intersecting at $P(6, 4)$ and both are tangent to x -axis and line $y = mx$ (where $m > 0$). If product of radii of the circles $S_1 = 0$ and $S_2 = 0$ is $\frac{52}{3}$, find the value of m^2 .
6. Consider the set of complex number z satisfying $|1 + z + z^2| = 4$. The maximum value of the imaginary part of z can be written in the form $\sqrt{\frac{m}{n}}$, where m and n are relatively prime positive integers. What is $m + n$?
7. In the following list of numbers, the integer n appears n times in the list for $1 \leq n \leq 200$.
1, 2, 2, 3, 3, 3, 4, 4, 4, 4, ..., 200, 200, ..., 200
What is the median of the numbers in this list?
8. Tetrahedron $ABCD$ has $AB = 5, AC = 3, BC = 4, BD = 4, AD = 3$, and $CD = \frac{12}{5}\sqrt{2}$. What is the volume of the tetrahedron?



SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

Partial Marks : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

9. Suppose a class has 9 boys & 1 girl. The average marks of these students in the mathematics exam is 60 & their variance is 82.2. A student fails in the exam if he/she gets less than 50 marks. Given that highest marks = 83 which are secured by the girl & the marks of no 2 students are the same & all marks $\in N$. Worst case is when maximum no. of students fail.
- A) In the worst case, the no. of students that can fail is 2
- B) In the worst case, the minimum marks a student can get is 48
- C) In the worst case, the average of 3rd best & 7th best is 61
- D) In the worst case, the standard deviation of the boys that have passed = 2
10. Let A_n be the area of region bounded by the curve $(2ny^2 - x)(y - x) = 0; n \in N$, then
- A) $A_1, A_2, A_3, \dots$, are in H.P.
- B) $\sqrt{A_1}, \sqrt{A_2}, \sqrt{A_3}, \dots$, are in G.P.
- C) $\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n \sqrt{A_r \cdot A_{r+1} \cdot A_{r+2}} \right) = \frac{1}{192\sqrt{6}}$
- D) $\lim_{n \rightarrow \infty} \left(\sum_{r=1}^n e^{\frac{1}{\sqrt{A_r}}} \right) = \left(\frac{1}{e^{2\sqrt{6}} - 1} \right)$
11. The graph of $y = \sqrt{x+3}$ and $y = \sqrt{1-x} + f(k)$ intersect, where k is real parameter
- A) if $f(k) = k^2 - 7k + 12$, then maximum value of k is 5
- B) if $f(k) = k$, then number of integral values of k is 5
- C) if $f(k) = k^2 - 7k + 12$, then minimum value of k is 2
- D) if $f(k) = k$, then number of integral values of k is 2





12. If $z = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$ and $\alpha = \frac{z}{z^2 + 1}, \beta = \frac{z^2}{z^4 + 1}, \gamma = \frac{z^3}{z^6 + 1}$. Choose the **CORRECT**

option(s)

A) $\alpha + \beta + \gamma = -2$

B) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -1$

C) β divides the line joining α and γ internally

D) $|\beta| < |\alpha| < |\gamma|$

13. Let $g(x)$ is a real valued function which is continuous function on $[0, 2]$ and differentiable in $(0, 1)$, then which of the following is / are always correct?

A) $\lim_{x \rightarrow 1} g(x)$ exist

B) $\lim_{x \rightarrow 1} \left[\frac{3}{2} g(x) \right]$ exist, when $[A]$ denotes greatest integer less than or equal to A

C) $|g(b)| - \left| g\left(\frac{1}{10}\right) \right| \leq \int_0^1 |g'(x)| dx \forall b \in (0, 1)$

D) $|g(b)| - \left| g\left(\frac{1}{10}\right) \right| \leq \int_0^1 |g'(x)| dx$ for atleast one value of b in $(0, 1)$

14. Let $f(x) = 2014 \tan^{2015} x + 2014 \tan^{2013} x - 2010 \tan^{2011} x - 2010 \tan^{2009} x$, for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Then the correct expression(s) is/ (are)

A) $\int_0^{\pi/4} x f(x) dx = \frac{1}{2011.2013}$

B) $\int_0^{\pi/4} f(x) dx = 0$

C) $\int_0^{\pi/4} x f(x) dx = \frac{2}{2011.2013}$

D) $\int_0^{\pi/4} f(x) dx = 1$



**JEE MAIN
2023**
SIRARAU
VENKAT KOUNDINNYA
SRI CHAITANYA
IIT Academy
RANK 1
300
300



**RANK
1**

**JEE Advanced
2023**
VARUN
CHIVILAS REDDY
SRI CHAITANYA
IIT Academy
RANK 1
341
360



**RANK
1**

**NEET
2023**
BODA VARUN
CHAKRAVARTHI
SRI CHAITANYA
IIT Academy
RANK 1
720
720



**RANK
1**



SECTION – III (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

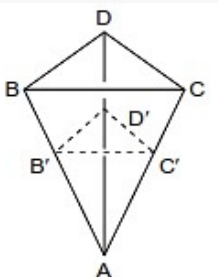
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 ONLY if the option corresponding to the correct combination is chosen;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -1 In all other cases.

15. A tetrahedron vessel has vertices $ABCD$ such that $\overrightarrow{AB} = \hat{i}$, $\overrightarrow{AC} = \hat{i} + \hat{j}$, $\overrightarrow{AD} = \lambda\hat{i} + \hat{j} + \hat{k}$ and it is completely filled with water



	List-I		List-II
(I)	If the water evaporates from the top surface at the rate proportional to height of water level from the bottom (vertex A)	(P)	$\frac{19}{27}$ of tetrahedron when $\lambda = 3$
(II)	If the water evaporates from the top surface at positive rate proportional to area of water exposed to air	(Q)	time to empty $\frac{7}{8}$ of total volume is $\frac{3}{4}$ th of the time to empty whole of the vessel
(III)	If the tetrahedron is cut through a parallel plane such that area of $\Delta B'C'D'$ is $\frac{4}{9}$ of area of ΔBCD then volume of figure $BDCB'D'C'$ is	(R)	time to empty $\frac{7}{8}$ of the total volume is $\frac{1}{2}$ of the time to empty whole of the vessel
(IV)	For $\lambda = 2$ the volume of the tetrahedron is	(S)	$\frac{27}{8}$ of tetrahedron when $\lambda = 3$
		(T)	same as of the tetrahedron having $\lambda = 1$

The correct option is:

- A) (I)-(R); (II)-(S); (III)-(T); (IV)-(P) B) (I)-(Q); (II)-(R); (III)-(P); (IV)-(T)
C) (I)-(R); (II)-(Q); (III)-(P); (IV)-(S) D) (I)-(Q); (II)-(R); (III)-(S); (IV)-(T)



JEE MAIN 2023

SINHA RAJ
VENKAT KOUNDINNYA
Sri Chaitanya
JEE Main Class
300
300

RANK

1
1

JEE Advanced 2023

VARSHIL
CHIDVILAS REDDY
Sri Chaitanya
JEE Advanced Class
341
360

RANK

1
1

NEET 2023

SODA VARUN
CHAKRAVARTHI
Sri Chaitanya
NEET Class
720
720

RANK

1
1



16. $f(x)$ is cubic polynomial with real coefficient and its leading coefficient = 1 and $g(x)$ is continuous function on R such that $f(x) \cdot g(x) = (x^2 + 3x + 2)$ and $g(-1) = 1$

	List-I		List-II
(I)	If $f(0)$ is integer, then	(P)	when $f(0)$ also takes maximum integral value is $-\frac{1}{3}$
(II)	If $f'(0)$ is integer, then	(Q)	the number of integral values of $f(0)$ is 7
(III)	The maximum value of $g(x)$	(R)	when $f(0)$ also takes maximum integral value is 1
(IV)	The minimum value of $g(x)$	(S)	when $f(0)$ also takes minimum integral value is $-\frac{1}{8}$
		(T)	number of integral value of $f(0)$ is 3

The correct option is:

- A) (I)-(T); (II)-(Q); (III)-(R); (IV)-(P) B) (I)-(Q); (II)-(T); (III)-(R); (IV)-(P)
C) (I)-(T); (II)-(Q); (III)-(P); (IV)-(S) D) (I)-(Q); (II)-(T); (III)-(P); (IV)-(R)

17. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(I)	Let p, q be real numbers such that $p^2 + q > q^2 + p$. Then value $p^2 + q$ is greater than	(P)	3
(II)	If equation $2t^3 - 9t^2 + 30 - a = 0$ has three real and distinct roots, then possible integral values of a is/are	(Q)	5
(III)	Let $f: [3, \infty) \rightarrow [1, \infty)$ be defined by $f(x) = \pi^{x(x-3)}$. Number of solutions of $f(x) = f^{-1}(x)$ is always greater than	(R)	$-\frac{1}{2}$
(IV)	Sum of series $\sum_{r=1}^{\infty} \sin^{-1} \left[\frac{2r+1}{r(r+1)(\sqrt{r^2+2r} + \sqrt{r^2-1})} \right]$ is always less than	(S)	-1
		(T)	8





The correct option is:

- A) (I)-(P, Q, R);(II)-(Q, S);(III)-(P);(IV)-(S, T)
 B) (I)- (P, Q, R);(II)-(P, R);(III)-(T);(IV)-(Q, T)
 C) (I)- (P, Q, R, S);(II)-(P, Q, R);(III)-(T);(IV)-(P, Q, R, T)
 D) (I)-(R, S);(II)-(Q, T);(III)-(R, S);(IV)-(P, Q, T)

18. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(I)	If $\lim_{n \rightarrow \infty} \left(n^{\frac{1}{2} \left(1 + \frac{1}{n} \right)} \left(1^1 2^2 3^3 4^4 \dots n^n \right)^{\frac{1}{n^2}} \right) = e^{-\left(\frac{p}{q} \right)}$ then the value of p/q is greater than	(P)	5
(II)	If $(7!)!$ is divisible by $(7!)^{k!} \cdot (6!)$ then the maximum value of k is	(Q)	6
(III)	The value of $\lim_{x \rightarrow 0} x^{(x^n - 1)}$ is less than or equal to	(R)	$\frac{1}{5}$
(IV)	If α, β, γ are the roots of the equation $x^3 - 2x^2 - 1 = 0$ and $T_n = \alpha^n + \beta^n + \gamma^n$ then the value of $\frac{T_{11} - T_8}{T_{10}}$ is greater than	(S)	$\frac{1}{6}$
		(T)	$\frac{1}{4}$

The correct option is:

- A) (I)-(R, S); (II)-(Q, S);(III)-(P); (IV)-(S, T)
 B) (I)-(P, S); (II)-(Q);(III)-(P, Q); (IV)-(R, S, T)
 C) (I)-(R); (II)-(Q, S);(III)-(S);(IV)-(P, T)
 D) (I)-(R, S); (II)-(Q);(III)-(P, Q); (IV)-(R, S, T)



PHYSICS

Max Marks: 60

SECTION – I
(NUMERICAL VALUE TYPE)

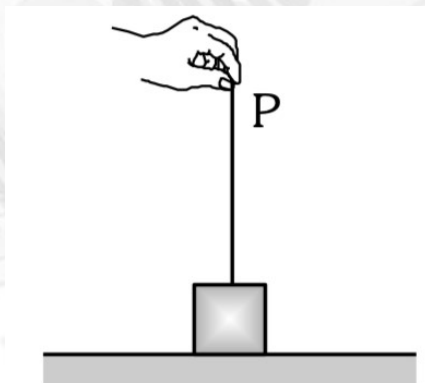
This section contains EIGHT (08) questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 **ONLY** if the correct numerical value is entered ;

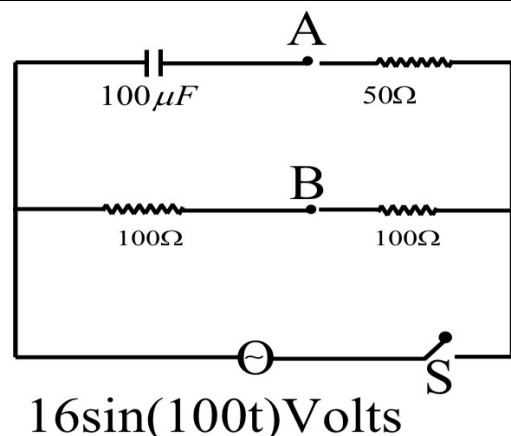
Partial Mark: 0 In all other cases.

19. One end of an elastic string is attached to a block of mass m and the other end is held keeping the string relaxed in vertical. Relaxed length of the string is l_0 and there is a mark P at height $0.8l_0$ from the lower end. Now the upper end is slowly raised. When the block leaves the ground, the mark P reaches the point where upper end of the relaxed string was. Expression for the work done by the force applied by the hand is $nmg l_0$. Then value of n is _____

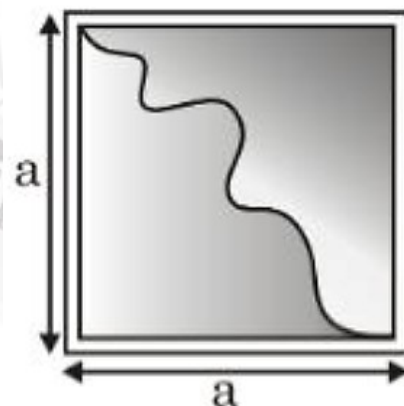


20. A string fixed at both ends is vibrating in the lowest mode of vibration for which a point at quarter of its length from one end is a point of maximum displacement. The frequency vibration in this mode is 100 Hz. The frequency emitted when it vibrates in the next mode such that this point is again a point of maximum displacement is f . The value of f is _____
21. ${}^7_3\text{Li} + {}^1_1\text{H} \longrightarrow {}^7_4\text{Be} + {}^1_0\text{n}$ is endothermic. What is the minimum kinetic energy (in keV) of incident proton so that the reaction occurs? The Q value of the reaction is -1645 keV
22. In the circuit shown in figure, find the maximum value of $(V_A - V_B)$ (in volts) after closing the switch in steady state.





23. The pitch of screw gauge is 1 mm and there are 50 divisions on its cap. When nothing is put in between the studs, 44th division of the circular scale coincides with the reference line, zero of the circular scale is lying above the reference line. When the glass plate is placed between the studs the main scale reads three divisions and the circular scale reads 26 divisions. Calculate thickness of plate (in mm).
24. A thin soapy film is stretched at the square frame with a length $a = 5.0$ cm. A slender flexible thread with negligible mass is fastened at two diagonal points of the frame, separating soapy film in two regions as shown at the picture below. The film on one side is pierced and the thread takes a circular shape. The length of the thread is $\pi a/2$ and it is inextensible. Find the tension (in mN) in the thread if the surface tension of the soap film is 3 mN/metre.

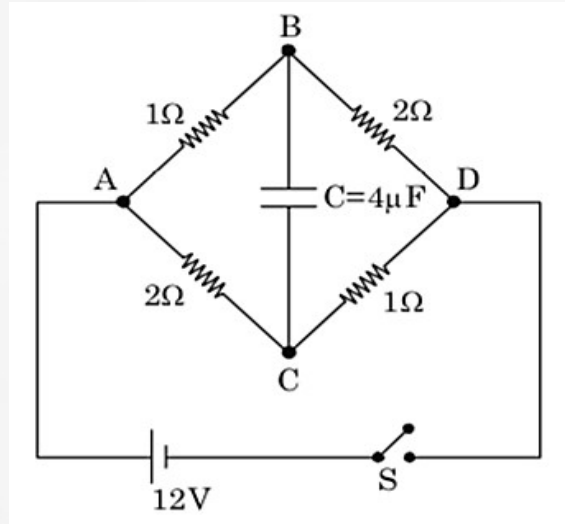




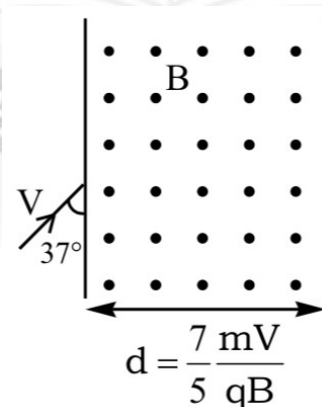
25. An uncharged capacitor of capacitance C is connected in given circuit diagram, switch S is closed at $t = 0$. If current in branch BC as function of time is given by

$$I = I_0 (\text{amp}) e^{-\frac{t}{\tau (\text{in } \mu\text{S})}}$$

then find the numerical value of $I_0 \tau$.



26. A particle of positive charge ' q ' and mass ' m ' enters a region of uniform magnetic field B with velocity V making an angle of 37° with the boundary of region. Magnetic field is directed outside the plane as shown in figure. The width of the region is $d = \frac{7mv}{5qB}$. The distance travelled by the particle in the region of magnetic field is $\frac{x(mV)}{y(qB)}$, here y is a prime number. Calculate $x + y$.



SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

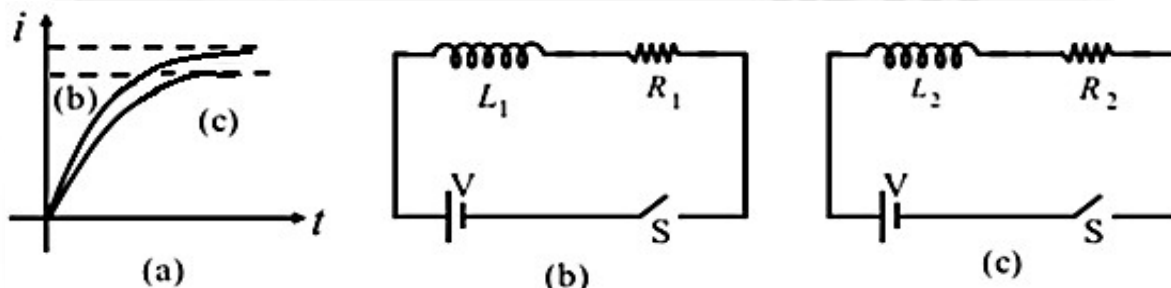
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

Partial Marks : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

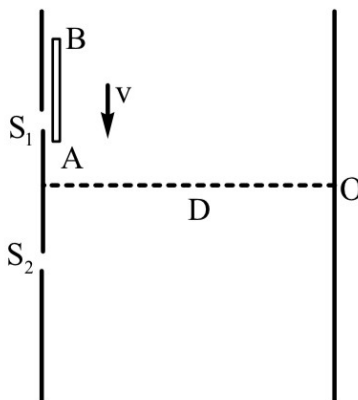
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

27. Current growth in two L-R circuits (b) and (c) is shown in figure (a). Let L_1, L_2, R_1 and R_2 be the corresponding values in two circuits. Then

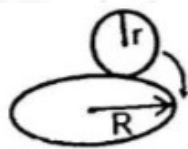


- A) $R_1 > R_2$ B) $R_1 = R_2$ C) $L_1 > L_2$ D) $L_1 < L_2$
28. In YDSE the separation between slits S_1 and S_2 is 5mm and slits are illuminating with light of wavelength $5000\text{\AA}$. The interference fringes are formed on the screen with central maximum at 'O'. A glass plate of $\mu = \frac{3}{2}$ with thickness $1\mu\text{m}$ at end A and is linearly increasing to $7\mu\text{m}$ at end B is moved in front of slits with constant speed 1mm/sec downwards. At $t=0$, end A is right in front of S_1 . $AB=6\text{mm}$





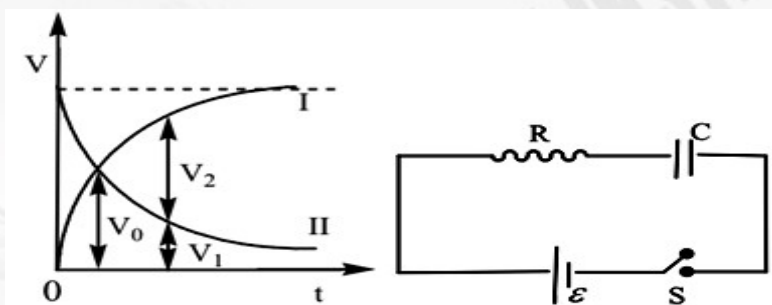
- A) At $t=0$, just after end A of plate comes in front of slit 1, the order of maxima at O is 1.
 B) At $t = 5 \text{ sec}$ 5th bright fringe is at O
 C) Order of maxima at O when B is in front of slit 1 is 5.
 D) Order of maxima at O when B is in front of slit 1 is 7.
29. A hypothetical pulse is travelling along positive x direction on a taught string. The speed of pulse is 10 cm/s and shape of pulse at $t = 0$ is given as
- $$y = \frac{x}{6} + 1 \text{ for } -6 \leq x \leq 0$$
- $$= -x + 1 \text{ for } 0 \leq x < 1$$
- $$= 0 \text{ for all other value of } x$$
- If x and y are in cm, then
- A) Vertical displacement of the particle at $x = 1 \text{ cm}$ at $t = 0.2 \text{ s}$ is $\frac{5}{6} \text{ cm}$
 B) Vertical displacement of the particle at $x = 1 \text{ cm}$ at $t = 0.2 \text{ s}$ is $\frac{4}{3} \text{ cm}$
 C) Transverse velocity of the particle at $x = 1 \text{ cm}$ at $t = 0.2 \text{ s}$ is $-\frac{5}{3} \text{ cm/s}$
 D) Transverse velocity of the particle at $x = 1 \text{ cm}$ at $t = 0.2 \text{ s}$ is $\frac{4}{3} \text{ cm/s}$
30. A ring of radius r is moving without slipping on a circular track of radius R . Angular speed of the ring about its axis is ω . Ignoring the fact that the ring axis is inclined.



- A) the angular acceleration of the ring is $\frac{\omega^2 r}{R}$
 B) the angular acceleration of the ring is $\frac{\omega^2 (R-r)}{R}$
 C) the angular velocity of centre of the ring about vertical axis passing through the centre of track is $\frac{r\omega}{R}$
 D) the angular velocity of centre of the ring about vertical axis passing through the centre of track is $\frac{(R-r)\omega}{R}$



31. A satellite is going round a planet in circular orbit of radius ' r '. Its kinetic energy is made double instantaneously, the direction of motion at that instant being unaltered. Then
- A) The satellite will describe elliptical motion with minimum and maximum distances from the planet being r and $r\sqrt{2}$
- B) The satellite will describe circular orbit of radius $r\sqrt{2}$
- C) The angular momentum of the satellite about the planet is increased by a factor $\sqrt{2}$
- D) The satellite will leave the gravitational field of the planet.
32. In the given $R - C$ circuit switch S is closed at $t = 0$ and the variation of potential drop across the resistor (V_R) and across capacitor (V_C) is represented by given plots, then



- A) $2V_0 = 2V_1 + V_2$
- B) $2V_0 = V_1 + V_2$
- C) curve I is for V_C and curve II is for V_R
- D) curve I is for V_R and curve II is for V_C

SECTION – III (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

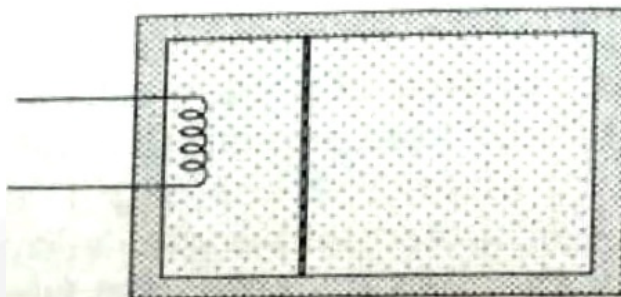
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: +3 ONLY if the option corresponding to the correct combination is chosen;

Zero Marks: 0 if none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -1 in all other cases.

33. A cylinder of volume $V = 10^{-2} \text{ m}^3$ made by non conducting walls contains 8 gm of He gas. The cylinder is divided by a thin fixed membrane into two unequal parts. Volume of the left compartment is $V/3$ and the right component has volume $\frac{2V}{3}$.



The left compartment has an installed electric heater. Due to finite conductivity of the membrane, heat passes from the left part into the right part. Heat transfer rate through the membrane is 0.2 watt per $\Delta T = 1^\circ\text{C}$ temperature difference across the membrane.

Initially the heater is off and both the parts are in thermal and mechanical equilibrium.

The membrane can withstand maximum pressure difference of 10^3 Pa . The heater starts supplying heat slowly the left compartment at its maximum possible power such that the membrane remains intact (after long time of heating also). (Take $R = 25/3 \text{ J/mol-K}$).

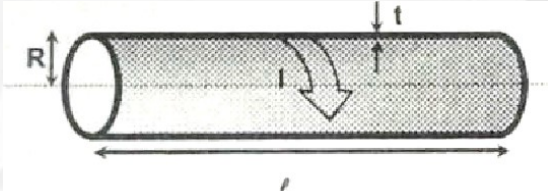
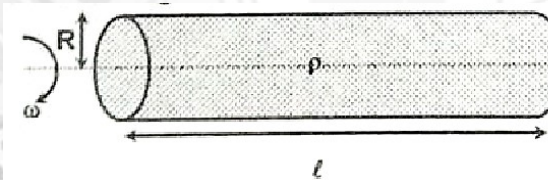
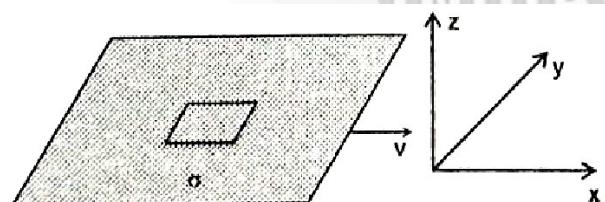
	Column-I		Column-II
(I)	For the left compartment	(p)	
(II)	For the right compartment	(q)	Rate of transfer of heat is 0.20 watt
(III)	For the whole cylinder	(r)	Rate of transfer of the heat is 0.06 watt
(IV)	Through the membrane	(s)	Rate of transfer of heat is 0.12 watt

The correct option is:

- A) (I)-(R); (II)-(S);(III)-(S); (IV)-(P) B) (I)-(R); (II)-(S); (III)-(P); (IV)-(S)
C) (I)-(S); (II)-(R);(III)-(P); (IV)-(S) D) (I)-(S); (II)-(R); (III)-(S); (IV)-(P)



34. Match the following

	Column-I		Column-II
(A)	A conducting loop is placed on horizontal xy plane in a uniform vertical magnetic field $\vec{B} = \frac{B_0}{t^2} T^2 (\hat{k})$ where T is a constant. The radius of the loop is increasing continuously as $r = \frac{r_0 t}{\Delta T}$, where t is starting from a value $t \rightarrow 0$	(p)	An induced current
(B)	A fixed long conducting cylinder of length ℓ, radius R and thickness t (where $\ell \gg R$ and where $\ell \gg t$) having an initial current I_0. 	(q)	An electrostatic field
(C)	A non conducting solid cylinder of length ℓ, radius R ($R \ll \ell$) having uniform volume charge density ρ rotating about its central axis with constant angular velocity ω by some external agent. 	(r)	An induced electric field
(D)	An infinite sheet of surface charge density σ placed on xy plane and moving with velocity $\vec{v} = v_0 \hat{i}$ and a conducting loop is fixed just above the sheet. 	(s)	A steady magnetic field

Sec: Sr.Super60_Elite, Target & LIIT-BTs

Space for rough work

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THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

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JEE 2023
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300



RANK 1

JEE Advanced 2023

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CHIVILAS REDDY
SRI CHAITANYA
JEE 2023
341
360



RANK 1

NEET 2023

BODHA VARUN
CHAKRAVARTHI
SRI CHAITANYA
NEET 2023
720
720



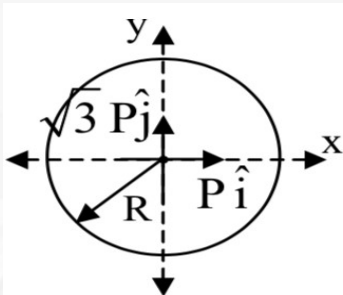
RANK 1



The correct option is:

- A) (I)-(S); (II)-(R);(III)-(Q); (IV)-(Q, S)
 B) (I)-(R); (II)-(S);(III)-(Q); (IV)-(P)
 C) (I)-(R); (II)-(S);(III)-(Q,S); (IV)-(Q, S)
 D) (I)-(S); (II)-(Q);(III)-(S);(IV)-(Q)

35. Column I gives a situation in which two dipoles of dipole moment $p\hat{i}$ and $\sqrt{3}p\hat{j}$ are placed at origin. A circle of radius R with center at the origin is drawn as shown in figure. Column II gives coordinates of certain positions on the circle. Match the statements in column I with the statements in column II.



	Column-I		Column-II
(A)	The coordinate(s) of point on circle where potential is maximum.	(p)	$\left(\frac{R}{2}, \frac{\sqrt{3}R}{2}\right)$
(B)	The coordinate(s) of point on circle where potential is zero.	(q)	$\left(-\frac{R}{2}, -\frac{\sqrt{3}R}{2}\right)$
(C)	The coordinate(s) of point on circle where magnitude of electric field intensity is $\frac{1}{4\pi\epsilon_0} \frac{4p}{R^3}$	(r)	$\left(-\frac{\sqrt{3}R}{2}, \frac{R}{2}\right)$
(D)	The coordinate(s) of point on circle where magnitude of electric field intensity is $\frac{1}{4\pi\epsilon_0} \frac{2p}{R^3}$	(s)	$\left(\frac{\sqrt{3}R}{2}, -\frac{R}{2}\right)$

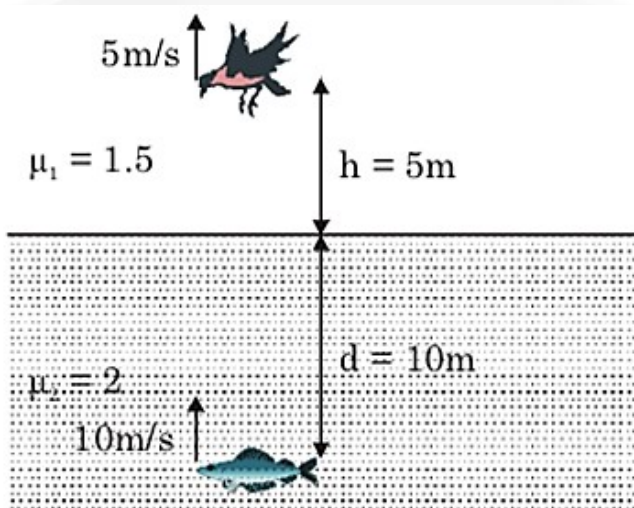
The correct option is:

- A) (I)-(P); (II)-(R, S); (III)-(P, Q); (IV)-(R, S)
 B) (I)-(P); (II)-(P, Q); (III)-(R, S); (IV)-(R, S)
 C) (I)-(R, S); (II)-(P); (III)-(R, S); (IV)-(P, Q)
 D) (I)-(P, Q);(II)-(R, S); (III)-(R, S); (IV)-(P)





36. Diagram shows a bird B and a fish F moving in two different medium of refractive index 1.5 and 2 with speed 10 m/s and 5 m/s respectively. At the given instant bird is located at height 5m above the interface while fish is located at depth of 10 m. List-I indicates various parameters while List-II gives its values. Match them correctly :-



	Column-I		Column-II
(A)	Separation of fish as observed by bird	(p)	$\frac{50}{3}$ unit
(B)	Separation of bird as observed by fish	(q)	12.5 unit
(C)	Speed of fish as observed by bird	(r)	$\frac{10}{3}$ unit
(D)	Speed of bird as observed by fish	(s)	2.5 unit

The correct option is:

- A) (I)-(R); (II)-(Q) ; (III)-(P);(IV)-(S)
 B) (I)-(Q); (II)-(P) ; (III)-(S);(IV)-(R)
 C) (I)-(S); (II)-(P) ; (III)-(Q);(IV)-(R)
 D) (I)-(P); (II)-(Q); (III)-(R);(IV)-(S)



CHEMISTRY

Max Marks: 60

SECTION – I
(NUMERICAL VALUE TYPE)

This section contains EIGHT (08) questions.

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme :

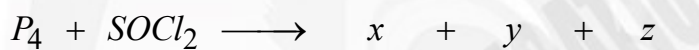
Full Marks: +3 **ONLY** if the correct numerical value is entered ;

Partial Mark: 0 In all other cases.

37. During extraction of iron, CO is the chief reducing agent. If 72 Kg of carbon is available and produced CO is completely used to reduce Haematite (present in excess amount), then calculate how much Kg of Iron is obtained after reduction.

(Given : atomic weight of C – 12 g / mol, O – 16 g / mol, Fe – 55.85 g / mol)

38. Consider the following incomplete and unbalanced chemical reaction:



X = active species in the HVZ reaction

Y = gas used to decolorise the magenta colour of para rosaniline hydrochloride

Z = reacts with ethane to form Yprite (mustard gas)

The sum of stoichiometric coefficients of X, Y and Z when the above reaction is balanced using simplest natural numbers is _____.

39. In a molecule of tetraborane (B_4H_{10}), if the number of $2c - 2e$ bonds is 'p' and the number of $3c - 2e$ bonds is 'q', report the value of $\frac{p}{q}$ (p divided by q)

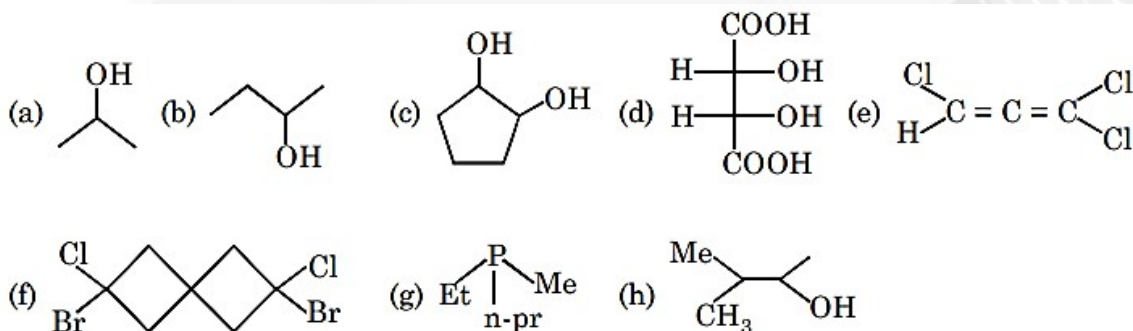
40. 1 mole each of acetic acid and ethyl alcohol are mixed at 25°C . When equilibrium is attained, 12 g of water is formed. What weight of ester (in grams) will be formed if the reaction is started with 2 moles each of acetic acid and ethyl alcohol at the same temperature.

41. 1000g of 1 molal aqueous solution of sucrose is cooled and maintained at -3.72°C . Calculate the weight of ice (in grams) that will separate out.

(K_f for water = 1.86°C/molal)

42. Ionic solid $AB(s)$ (M. wt. = 50) crystallise in $NaCl(s)$ type structure. If edge length of unit cell is $\frac{1}{3} \times 10^{-7}$ cm, then calculate density of $AB(s)$ in gm / ml if it contain $\frac{100}{3}\%$ schottky defect $[N_A = 6 \times 10^{23}]$

43. Which of following molecule shows optical isomerism



44. The pressure of H_2 that would be required to make the EMF of a hydrogen electrode zero when immersed in pure water at $25^\circ C$ is 10^x bar. The value of x is _____

SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +4 ONLY if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

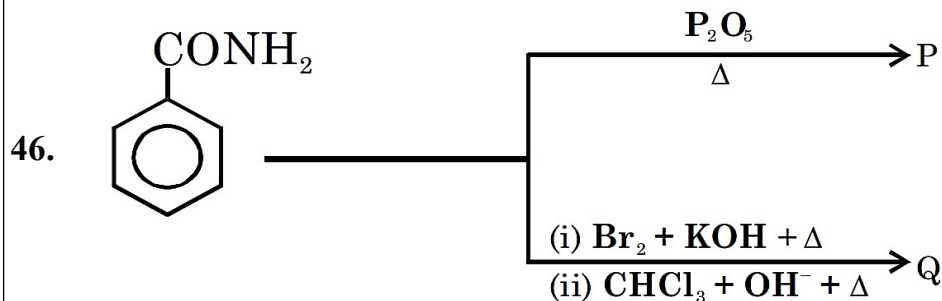
Partial Marks : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

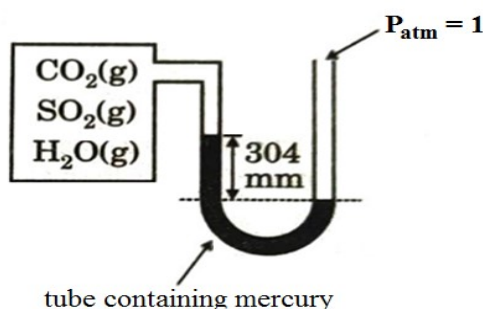
45. Which statement(s) is/are correct -

- A) If $0.1M$ CH_3COOH aqueous solution ($K_a = 1.8 \times 10^{-5}$) is diluted at $25^\circ C$, then $[H^+]$ will increase.
- B) On heating a sample of pure water $[H^+]$ and $[OH^-]$ increases but the sample still remains neutral
- C) $10^{-4}M$ HCl solution is less acidic than $0.1M$ HCN ($K_a_{HCN} = 10^{-5}$) solution
- D) $10^{-8}M$ HCl aqueous solution will have pH exactly equal to 8



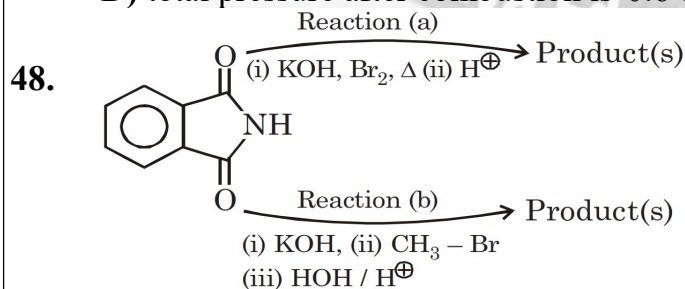
Which statement/s is/are CORRECT:

- A) Reduced product of P and Q will be metamers to each other.
- B) Dry distillation of hydrolysed product of P with $Ca(OH)_2$, gives benzophenone.
- C) Hydrolysed benzenoid product of Q, reacts with $NaNO_2 + HCl$ followed by reaction with phenol in alkaline medium, gives orange red dye.
- D) Electrophile involved in the formation of Q in step (ii) is dichlorocarbene.
47. A 82.1 L container connected with manometer contains mixture of CS_2 and H_2S gases and just the required amount of O_2 is added to form CO_2, SO_2 and $H_2O(g)$ at $227^\circ C$. Final condition of manometer is shown. (atmospheric pressure = 760 mm Hg, assume ideal gas behavior)



If moles of SO_2 gas produced is 0.7, select the correct statement(s)

- A) moles of CS_2 originally present is 0.3
- B) moles of CS_2 originally present is 0.2
- C) moles of H_2S originally present is 0.3
- D) total pressure after combustion is 0.6 atm



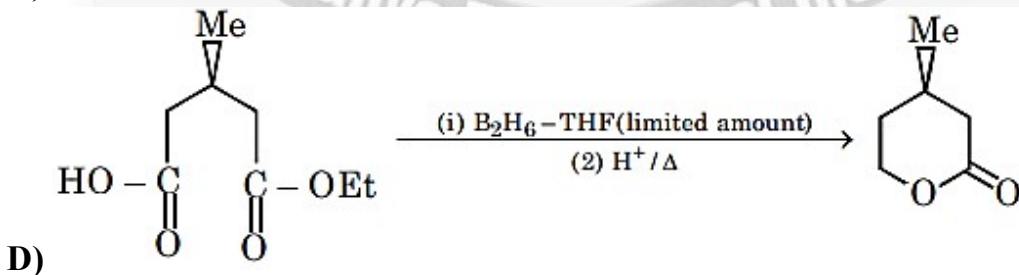
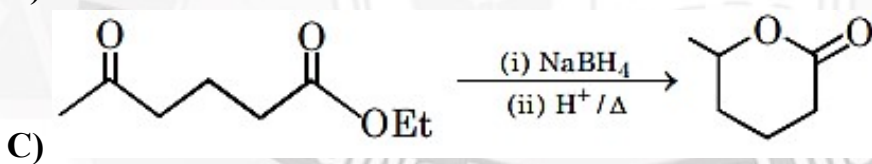
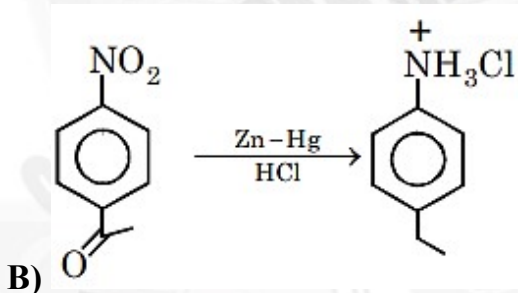
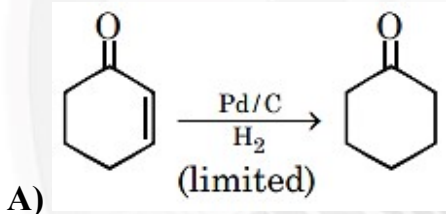
Correct statement about reaction (a) and (b) is / are

- A) Product(s) of reaction (a) gives isocyanide test
- B) Product(s) of reaction (b) gives isocyanide test
- C) Products of reaction (a) gives coupling reaction and also dye test with benzene diazonium chloride
- D) Product of reaction (b) gives coupling reaction and also dye test with benzene diazonium chloride

49. Select CORRECT statement(s).

- A) Blue solution of Na in liq. NH_3 is unstable with respect to amide formation
- B) Na_2O_2 react with CO and evolve O_2
- C) Li^+ does not form alum due to its high hydration energy
- D) NaNO_3 when heated at 800°C produce two gases both are paramagnetic

50. Identify reactions correctly matched with their major product?



SECTION – III (MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks: +3 ONLY if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -1 In all other cases.

51.

Column-I Ionic solid		Column-II Tetrahedral void occupancy	
P	Wurtzite	1	Vacant
Q	CaO	2	Ca^{2+}
R	Chromite	3	Fe^{2+}
S	Fe_3O_4	4	Zn^{2+}
		5	Fe^{3+}

Codes:

- A) P-4; Q-2; R-1; S-3 B) P-4; Q-2; R-3; S-5
 C) P-1; Q-2; R-3; S-5 D) P-4; Q-1; R-3; S-5

52.

Column – I Name reaction		Column – II Product(s)	
P)	Perkin reaction	1	Aniline
Q)	Oppenauer oxidation	2	Acetophenone
R)	Reimer Tiemann reaction	3	Cinnamaldehyde
S)	Schmidt reaction	4	Salicylaldehyde
		5	not mentioned above

Codes:

- A) P-3; Q-2; R-4; S-1 B) P-3; Q-1; R-5; S-2
 C) P-3; Q-2,4; R-4; S-1 D) P-5; Q-2; R-4; S-1





53.

	Column I (Ligand abbreviation)		Column II (Description)
P	en	1.	neutral ligand
Q	gly	2.	anionic ligand
R	dipy	3.	symmetrical bidentate
S	acac	4.	unsymmetrical bidentate
		5.	forms six membered chelate ring with a metal centre

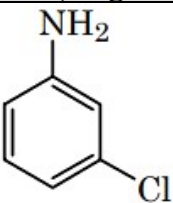
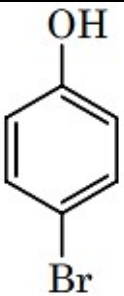
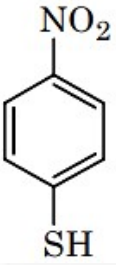
Codes:

A) P-1, 3; Q- 2, 4; R- 1, 3;S-2, 3, 5 B) P-1, 3; Q-2, 4; R-1, 3, 5;S-2, 4, 5

C) P-1, 3,5; Q-1, 4; R-1, 3;S-2, 3, 5 D) P-1, 3; Q- 2, 4, 5; R-1, 3;S-2, 3

54.

Matching the following

List-I (Organic compound)		List-II (Estimation/Detection Possible using)	
(P)		(1)	Dumas method
(Q)		(2)	Kjeldahl's method
(R)		(3)	Carius method
(S)	$\text{C}_2\text{H}_5\text{NH}_2$	(4)	$p\text{-CH}_3\text{C}_6\text{H}_4\text{SO}_2\text{Cl}$ (Hinsberg method)

Codes:

A) P-(1,2,3); Q- (3,4) ; R- (1,2); S- (1,2,4)

B) P- (1,2,3,4); Q- (3); R- (1,3); S- (1,2,4)

C) P- (1,2,3); Q- (1,4); R- (2,3); S- (4)

D) P- (1,4) ; Q- (3,4); R- (2,3); S- (1,3,4)

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BELOW 20
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Category Ranks

10
RANKS (50%)

BELOW 100
All India Open
Category Ranks

32

BELOW 1000
All India Open
Category Ranks

181

BELOW 100
All India
Category Ranks
Count

89

BELOW 1000
All India
Category Ranks
Count

699

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